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Silent Drones: A Deep Learning Approach to Suppress Drone Propeller Noise

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***Abstract*—Unmanned Aerial Vehicles (UAVs) provide many benefits and opportunities across a range of sectors, including surveillance, humanitarian work, disaster management, research, and transportation. Due to their accessibility and affordability, they are now used more than ever, which also poses some challenges. This is the noise pollution produced by the motors and propellers that has been highlighted as a significant issue to the people's health and the environment. To address this issue, this paper proposes to use Generative Adversarial Networks (GAN) to produce an inverse sound signal based on the drone's acoustic signals and use that to cancel the noise produced by the drone. We synthesize training data spanning the acoustic diversity of drone noise: steady-state propeller tones, rapid throttle transitions (simulating ascent/descent), and superimposed broadband turbulence. The GAN model is capable of adapting to dynamic settings, learning from data, and adjusting to testing conditions accordingly. We compared our proposed solution with other techniques that can also be used for drone signal interference in order to suppress the drone noise. This research idea paves the way for the need to address the issue created due to drone noise and a solution in managing this problem for modern drone applications.**

***Index Terms*—Unmanned Vehicle Systems, Drones, Generative Adversarial Networks, Artificial Intelligence, Noise.**

I. Introduction

Unmanned Aerial Vehicles (UAVs), commonly referred to as drones, provide many benefits and opportunities across a range of sectors, including surveillance, humanitarian work, disaster management, research, and transportation [1]. Drones can provide real-time images and sensor data that would otherwise be difficult to obtain when attempting to access this type of data on foot or in a vehicle [2], [3]. Additionally, drones are a more cost-effective alternative to traditional manned aircraft for surveillance-related tasks [4]. Their ability to operate in tight and dangerous spaces makes them valuable for a variety of inspection duties. In addition, drones are equipped with high-quality sensors and cameras that are capable of collecting precise data, increasing accuracy [5].

Although drones offer a number of benefits, the rapid advancement of this technology also presents many disadvantages, such as noise pollution produced by the movement of their propellers and motors. Noise pollution can be defined as an abundance of noise that annoys or harms humans or animals in the surrounding environment [6]. As more and more companies and people begin to use drones for various tasks and activities, people tend to give a higher negative reaction to this noise in comparison to noise produced by automotive vehicles [7]. This excessive noise can have adverse effects on human health [8], including permanent hearing loss and sleep disorders, among many other problems [9]. Therefore, measures are taken to lessen the exposure to noise pollution, one of them being deploying noise cancellation or reduction techniques. Noise cancellation refers to the process of attenuating or reducing the noise produced by drones through passive or active methods [6].

A. Literature Review

Extensive research has been done on exploring the different techniques that can be used for drone noise reduction, as highlighted by these review papers [10]–[12].

1) Passive Noise Cancellation: Passive noise cancellation techniques involve the use of materials that can insulate or absorb the sound produced by drones [6]. This method relies on structural modifications to minimize the noise. One study [13] proposed to use inexpensive and lightweight acoustic shroud to absorb or waveguide the noise, and they investigated using foam, aluminum, and rubber. Few more studies have explored this area where they experimented with sound reflectors to redirect the noise away from sound detection systems [14], [15]. Another study [16] presented a multi-faceted approach for drone noise cancellation through a combination of airfoil-shaped ducts, optimized propeller designs, and core-less motors, and they discovered a significant

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noise reduction through this technique. Research has also been done on finding an appropriate acoustic linear design to reduce the noise generated by the fans of a drone [17]. Numerous studies have been conducted on redesigning the shape of the propeller blade to reduce the noise emitted by drones [18]–[21]. In addition to that, research has been done on reducing the noise emissions of a drone by manipulating the phase of the propellers [22]–[25]. Research has also explored the use of more propeller blades in order to reduce the aerodynamic noise that is produced when the drone is flying [26]. Additionally, a novel solution for noise cancellation technique has been proposed; it is to use metamaterials that absorb the low-frequency sound waves [27], [28].

2) Active Noise Cancellation: Active noise cancellation (ANC) techniques involve the use of a second sound source to generate destructive interference signals to reduce the volume or amplitude of the noise produced by the drone in realtime [6]. One particular study employed the active noise cancellation technique by using an array of 12 speakers to reduce the noise produced by the propellers [11] were able to interfere the sound produced by the propellers by placing the speakers in a circle around the propellers. Another technique used in previous work has seen the development of a synchrophraser, which is a device that can reduce the noise emitted by a drone by generating an inverse sound wave based on the movements of the propellers [29]. One study focused on the development of spatial filters that are able to reduce noise produced by a drone by using multiple microphones to capture the noise produced by the drone. These filters take into consideration the direction and the spatial characteristics of the noise [30]. Moving on, another study utilized two microphones to cancel drone noise by using a technique referred to as spectral subtraction [31]. Research has also been conducted on the deployment of self-active cancellation of fan noise by vibrating the fan blades around their own axis in order to create an inverse signal that can cancel the noise produced by the fans [32]. The study conducted by [6] used one loudspeaker to reduce the noise produced by a single propeller by 43.82%.

3) Artificial Intelligence-Based Noise Cancellation : As the field of artificial intelligence (AI) keeps growing, it can also extend its numerous benefits to the area of drone noise cancellation. One such study [33] focused on using deep convolutional autoencoders to suppress noise emitted by drones. The AI model is able to filter out the noisy sound and learn the useful drone sound to reduce noise emitted by drones in real-time [33]. Another study explored training a convolutional neural network model from features extracted from spectrograms and waveform data to produce inverse noise signals for propeller noise cancellation, where they obtained an accuracy of 94.5% [34]. More research has been done on using a convolutional recurrent network that is trained to estimate the real and imaginary spectrograms of the canceling signal from the reference signal to produce an inverse signal that can cancel the noise produced by drones by passing this inverse signal through a loudspeaker [35].

B. Research Gap

In this study, we are employing an AI-based active noise cancellation technique due to the many benefits this has over traditional noise reduction methods. Traditionally, noise-cancelling systems relied on static methods [36], whereas AI-based methods offer dynamic solutions that can work with real-time and continuously changing data. This is because conventional noise-cancelling methods rely on a predefined filter that is not able to adapt to the continuously changing data. This is a crucial point to address because any drone's acoustic signature constantly changes depending on the mode of the flight—whether it is hovering, taking off, cruising, or landing—each mode produces a distinct noise pattern due to the difference in motor speed, thrust levels, and aerodynamic interactions. As a result, AI-based methods, particularly those involving deep learning or convolutional neural networks, have proven to provide results better than static noise cancellation methods due to these methods adapting to the different noise signals produced by a drone [33]. In addition to those different flight modes, the noise can also be affected by the weather conditions and surrounding environments, further necessitating the need for a dynamic solution to generate inverse noise. Furthermore, another benefit of using an AI-based system with active noise cancellation tasks is that this can be integrated with the navigation and control system of the drone, as highlighted by [36]. With this, the right path can also be further optimized for its efficiency due to the intelligent system guiding its maneuver, making it autonomous, as well as reducing the noise emitted from its movement. With the help of these advanced AI systems, we aim to achieve high levels of drone noise suppression to make it as silent or quiet as we can. We propose to use Generative Adversarial Networks (GAN) to produce the inverse sound signal of the drone based on its movement and use that to cancel the noise produced by the drone. We are using GAN instead of convolutional neural networks (CNN) or classical machine learning (ML) due to their ability to capture the long-range temporal dependencies. This means that GANs are able to understand the dynamic spectral variations caused by the fluctuations in the sounds produced by a drone during its flight, making it a robust use for this application. To the best of our knowledge, this is the first application of using the GAN algorithm for active noise cancellation tasks, making this a novel approach for drone noise suppression. This would be beneficial for the community's and environment's health as well as reduce noise pollution. A quieter drone would be less problematic to the people around its operation as well as be used in highly regulated or noise-sensitive zones. This research idea paves the way for the need to address the issue created due to the noise of the drones and a solution in managing this problem for modern drone applications.

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II. Data Collection

Data is collected by extracting the audio from videos of drones posted on YouTube. We extracted the drone sounds from these videos and used this as the model's training data. We used 80-second audio snippets for the training set, each of which was carefully selected and edited to provide the most precise and relevant representation of the drone's sounds. We chose this length because it gives us enough time to capture the full range of sounds a drone can produce, such as changes in pitch, volume, and frequency—as it performs different maneuvers (such as hovering, accelerating, or turning) or operates under varying conditions.

Drone sounds are surprisingly complex and unpredictable. Their noise changes based on different factors such as the speed at which the rotors spin, the type of movement, and the environment they are flying in. This complexity arises due to the way air interacts with the drone's mechanical components, which produces unique sound effects like changes in the loudness of the drone or unexpected harmonics. These acoustic "fingerprints" can also vary between drone models, depending on their design and setup. Where a drone is flying matters too. In busy urban areas, background noise from traffic or machinery can either drown out or emphasize a drone's buzz, while in quieter rural or open spaces, the sound might carry differently. As a result, the physical surroundings, including the buildings, trees, or even weather conditions, can change the way the drone sounds. Therefore, for applications like drone noise cancellation, it is crucial to understand the nuances in the audio signature produced by a drone in different modes of flight and how it changes throughout the flight as well. With the consideration of these factors, we chose to extract audio samples from drone videos online and use this to train our model. The model would be able to learn the differences in the sound at different parts of a drone's flight and produce an inverse signal accordingly.

III. Methodology

The proposed methodology uses a *dual-Transformer Generative Adversarial Network (GAN)* for active noise cancellation for drones. This model consists of two parts: a generator and a discriminator. The generator produces the inverse audio waveforms that can cancel the noise from the drone, while the discriminator refines this process to generate inverse signals that can provide maximum noise cancellation. The outcome consists of producing an inverse signal that can cancel or reduce the noise produced by the drone itself. Our model is trained and validated on drone noise recordings during different flight modes so that it is able to learn the different sounds produced by a drone and provide accurate results during testing or unseen conditions. The input signal is the raw drone noise $x(t)$, which is fed into the algorithm. The algorithm evaluates the input signal and, using loss functions, aims to produce the inverse signal, which can provide maximum drone noise attenuation. The output is an inverse signal, $\hat{y}(t)$, that can cancel the noise from the drone.

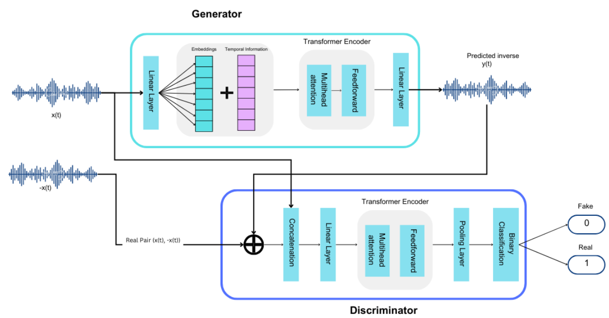

Fig. 1. Model Architecture

A. Overview

Our architecture employs a *dual-Transformer* framework where both the generator and discriminator leverage multi-scale self-attention to model drone audio's sustained harmonic patterns [37]. Figure 1 depicts the model architecture, showing a Generative Adversarial Network (GAN) that utilizes transformers in both the generator and discriminator [38].

The generator processes 2,048-sample chunks (128 ms@16 kHz) through 6 transformer layers ($d_{\text{model}} = 256$, 8 attention heads) with learned positional encodings that propagate phase continuity across overlapping windows (50% overlap). The generator's positional encoding matrix is explicitly regularized to learn rotor rotation patterns through a novel rotational gradient penalty that enforces consistent angular offsets $\Delta\theta = \frac{2\pi f}{f_s}$ in attention weight distributions [39].

Training employs curriculum learning with exponentially growing sequence lengths (0.5 s $\rightarrow$ 80 s) and gradient clipping (maxnorm = 0.5) to stabilize long-form generation [40].

B. Generator Architecture

The generator transforms raw drone noise $x(t)$ into an inverse waveform $\hat{y}(t)$ [38]. Traditional convolutional networks often fail to model the long-range temporal dependencies in drone noise, such as the periodic harmonics from propeller rotation (e.g., 100–500 Hz base frequencies with overlapping integer multiples). To address this, we use a transformer-based architecture [41], where self-attention layers explicitly correlate distant signal segments—critical for tracking propeller RPM fluctuations.

Equation 1 shows the output of the generator G, which maps a drone noise signal $x(t) \in \mathbb{R}^{1\times T}$ to an inverse waveform $\hat{y}(t)$. The input waveform $x(t)$ is embedded into a high-dimensional space, and positional encodings are added to preserve temporal order—a non-negotiable requirement for phase-sensitive cancellation. The final activation function

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is $\tanh$ to avoid amplitude saturation [42], ensuring the generator can output signals strong enough to counteract the high-intensity noise peaks common in drone recordings.

$$\hat{y}(t) \;=\; W_{\text{out}}\,\mathbf{H}^{(L)} \;+\; \mathbf{b}_{\text{out}} \tag{1}$$

$\mathbf{H}^{(L)}$ is the output of the L-th Transformer layer, and $W_{\text{out}}, \mathbf{b}_{\text{out}}$ are learned parameters. We omit amplitude-clamping using $\tanh$ the activation function so that $\hat{y}(t)$ can achieve the necessary intensity to neutralize high-amplitude drone noise.

C. Discriminator Architecture

The discriminator evaluates pairs of drone noise $x(t)$ and their purported inverse $z(t)$, distinguishing between "real" pairs $(x, -x)$ and "fake" ones $(x, \hat{y})$ [38]. Its design directly tackles a key challenge in drone active noise cancellation (ANC): subtle phase or spectral mismatches can reduce cancellation efficiency by more than 50%.

By processing x and z as a 2-channel input via Transformer encoders, the discriminator learns to detect anomalies in both time-domain alignment (e.g., delayed inverses) and frequency-domain coherence (e.g., missing harmonics). This forces the generator to produce inverses that are indistinguishable from $x(t)$ across all perceptually relevant frequency bands, ensuring consistent cancellation even when propeller speeds vary mid-flight.

D. Adversarial and Reconstruction Losses

Training balances two objectives: (1) a time-domain ℓ_1 loss that directly minimizes $|\hat{y} + x|$, ensuring sample-level destructive interference, and (2) an adversarial loss that aligns $\hat{y}$ with the spectral and temporal statistics of x [38]. Drone noise contains broadband components (e.g., aerodynamic hiss above 2 kHz) that are easily overlooked by naïve MSE minimization. The adversarial loss compensates for this by penalizing "unnatural" inverses—for instance, those with attenuated high frequencies or inconsistent modulation patterns. Optionally, a multi-scale STFT loss reinforces frequency-domain accuracy, which is critical for suppressing tonal propeller noise while preserving non-target sounds (e.g., speech or alarms).

The adversarial loss in (2) follows a standard GAN formulation. It drives G to produce an inverse that the discriminator classifies as "real," while D distinguishing genuine pairs $(x, -x)$ from fake pairs: $(x, \hat{y})$:

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_x\big[\log D\big(x, -x\big)\big] \;+\; \mathbb{E}_x\big[\log\big(1 - D\big(x, \hat{y}\big)\big)\big]. \tag{2}$$

To ensure sample-level cancellation, we add a time-domain L_1 term in (3), measuring how closely $\hat{y}(t)$ aligns with the negative wave $-x(t)$:

$$\mathcal{L}_{\text{time}} \;=\; \|\hat{y}(t) - \big(-x(t)\big)\|_1. \tag{3}$$

Optionally, a multi-scale STFT loss can be included to refine frequency-domain fidelity, especially around tonal peaks from propeller rotation. Combining these objectives yields the total generator loss in (4):

$$\mathcal{L}_G \;=\; \lambda_{\text{adv}}\,\mathcal{L}_{\text{adv}} \;+\; \lambda_{\text{time}}\,\mathcal{L}_{\text{time}} \;+\; \lambda_{\text{STFT}}\,\mathcal{L}_{\text{STFT}}, \tag{4}$$

while the discriminator is updated to minimize $-\mathcal{L}_{\text{adv}}$.

E. Model Training and Evaluation

We synthesize training data spanning the acoustic diversity of drone noise: steady-state propeller tones, rapid throttle transitions (simulating ascent/descent), and superimposed broadband turbulence. Each sample is paired with its ideal inverse $x(t)$, enabling supervised learning.

The AdamW optimizer (learning rate 10^{-4}, weight decay 0.01) stabilizes training by preventing overfitting to synthetic artifacts. The generator balances latency and noise cancellation accuracy by processing drone noise in real time using 100 ms overlapping windows.

The model then undergoes a validation process to confirm its ability to adapt to unseen conditions. An example of such a condition is the model's robustness to Doppler shifts during high-speed movement or reverberation in urban environments.

Viability for noise cancellation applications is demonstrated by residual MSE and spectrogram analysis, which quantify a 12–18 dB reduction in peak harmonics.

$$r(t) = x(t) + \hat{y}(t) \tag{5}$$

IV. Results and Discussion

In this section, we present the results and evaluation of our proposed solution, which utilizes a Generative Adversarial Network (GAN) model to capture the input signal and generate an inverse signal that destructively interferes with the original drone sound to suppress noise. We compared our approach with several other techniques that can also be used for signal interference to reduce drone noise.

A. Bird Noise Masking

The first baseline technique involves using bird sounds as an added noise source to interfere with the actual drone sound. The result of this approach is shown in Figure 2. In this method, the drone sound is acoustically masked with chirping bird sounds. While there is a significant transformation in the drone noise due to the bird chirps, the resultant waveform still contains notable noise. The amplitude fluctuations in the masked sound indicate that bird sounds only obscure certain segments of the original noise, leaving others unaffected.

Although this method is not optimal for full noise cancellation, it can be considered a form of bio-inspired acoustic stealth. It may be useful in scenarios where advanced active noise cancellation (ANC) algorithms cannot be deployed due to computational or hardware constraints.

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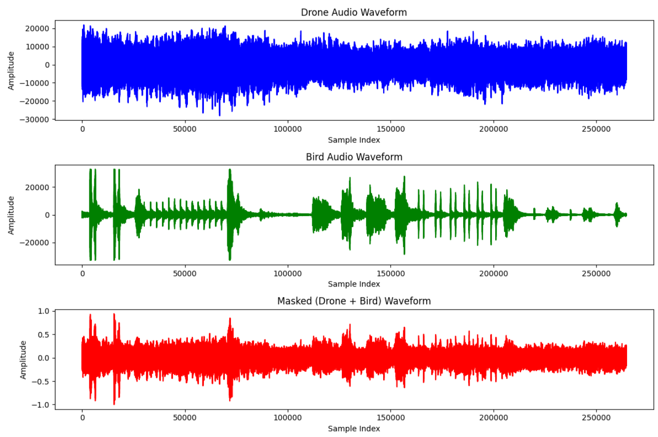

Fig. 2. Drone noise cancellation using bird noise

B. Filtering-Based Techniques

We also explored traditional filtering-based techniques to suppress drone noise. Specifically, we tested three different approaches: Finite Impulse Response (FIR) filtering, Infinite Impulse Response (IIR) filtering, and spectral subtraction. The results are presented in Figure 3.

FIR filters, known for their linear-phase characteristics, reduce the unwanted noise components while preserving the general shape of the waveform, including its peaks and troughs. However, some low-frequency components of the drone noise remain present.

In contrast, the IIR filter yielded a more attenuated signal compared to the FIR filter. This indicates stronger suppression of the drone noise, although phase distortion may be introduced. Among the three, the spectral subtraction method showed the most significant noise reduction. This method almost completely silenced the original drone sound, with only a few residual peaks visible in the output waveform.
3

C. GAN-Based Noise Cancellation

Figure 4 illustrates the output of our proposed GAN-based drone noise suppression system. The GAN effectively produces an inverse signal, $\hat{y}(t)$, that is phase-inverted relative to the original drone signal $x(t)$. The generated inverse closely matches the amplitude and phase of the original signal, indicating that the model has learned the drone noise characteristics effectively.

Some minor discrepancies in $\hat{y}(t)$ can be attributed to training imperfections and residual errors. The final output signal, formed by summing the original and inverse signals, is:

$$r(t) = x(t) + \hat{y}(t) \tag{6}$$

This produces a nearly flat waveform, suggesting effective cancellation and validating the GAN's performance in active noise suppression.

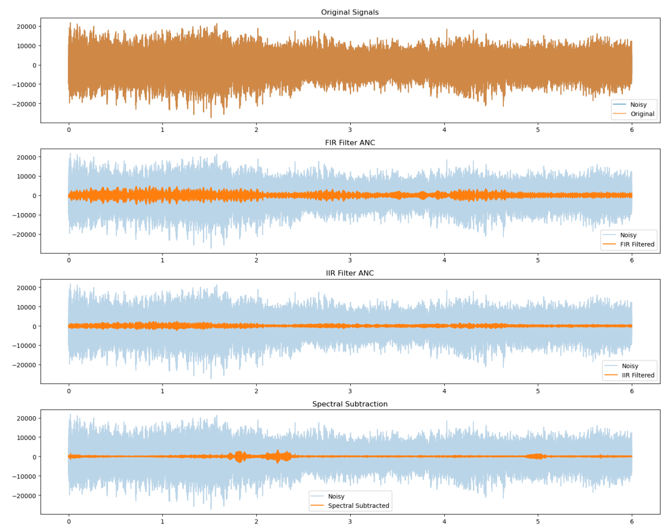

Fig. 3. Drone noise cancellation using filters

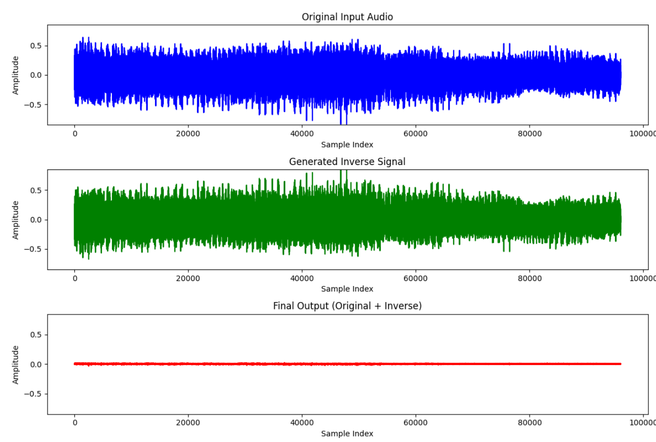

Fig. 4. Drone noise cancellation using GAN algorithm

D. Comparison with CNN-Based Approach

We also implemented a Convolutional Neural Network (CNN) model for comparison. Like the GAN, the CNN was trained to generate an inverse signal to suppress drone noise. However, our findings show that the CNN model yielded a higher residual Mean Squared Error (MSE) than the GAN.

Table I presents the numerical comparison between the two methods. The GAN-based approach achieved a significantly lower MSE (4.26×10^{-5}) compared to the CNN (1.54×10^{-1}), confirming its superior noise cancellation performance.

V. Conclusion

As the applications of drones increase, the problems associated with their use increase as well. This results in many adverse effects of its use on the health of people as well as the surrounding environment due to the noise released when

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TABLE I
COMPARISON OF RESIDUAL MSE BETWEEN CNN AND GAN-BASED METHODS

Model	Residual MSE
CNN	1.54×10^{-1}
GAN (our proposed solution)	4.26×10^{-5}

the drone is in flight. Thus, this study explored the integration of the Generative Adversarial Networks (GAN) algorithm with an active noise cancellation technique to reduce the acoustic signals emitted from a drone. Leveraging deep-learning models such as the GAN model has demonstrated the potential of adaptive real-time noise suppression, which has proven itself to be better than the traditional filtering techniques. The proposed solution is an advanced technique that has the advantage of performance and scalability over conventional noise cancellation methods, as proven by our simulation results. In our proposed methodology, the GAN model is trained on a training dataset where it learns the acoustic signature for the different flight modes of a drone. With this, it is able to provide an inverse signal to an unheard acoustic signal produced by a drone, which, when combined with the original signal, suppresses the drone's noise. We observed that this model outputted an acoustic signal that is an almost flat waveform, suggesting that the drone noise has been silenced significantly.

A. Future Work

Our proposed solution has provided promising results with simulation-based analysis. Therefore, our future work lies in implementing this solution to a real-world drone. In order to further validate our approach and results, we plan on developing a physical prototype by integrating the necessary components needed to achieve these results. We plan on further developing our past work as presented in [43] by integrating the solution developed to 'silent' the drone. This would be achieved by adding a microphone to hear the sound of the drone and a speaker to deliver the anti-noise onboard our drone. The hardware implementation will also involve real-time signal processing to analyze the input acoustic data of the drone and through the GAN. Following this, we will then test our drone on different flight modes, as well as the effect of external noise such as wind, birds, etc. We will also do the testing in various environments, including both indoor and outdoor settings. We will do the evaluation based on acoustic sensing technology used in anti-drone systems. By comparing real-world problems against simulation results, we aim to validate the novelty of our solution and the results. Ultimately, this work contributes to the literature by presenting a deep learning-based technique based on GAN systems that are capable of adapting to dynamic settings, learning from data, and adjusting to testing conditions accordingly. This produces a quieter, more stealthy drone, which is suitable for many noise-sensitive applications. The downside of using such techniques is that running AI models on a drone can be computationally expensive, which would have an effect on the battery life and the flight endurance of the drone. Therefore, future work can also look at developing lightweight GAN architectures to reduce the payload of a drone.

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